

## GENERAL

**ALIAMAG** The AMF601 Series is a sanitary-type electromagnetic flowmeter ideal for conductive liquids. It comes in sizes from 10-80 mm. The AMF601 is widely used for drink water, milk, food & beverage juice and many other industrial fluids.

AMF601 Series electromagnetic flowmeter can be used in compact or remote model with AMC Series converter of electromagnetic flowmeter.

## FEATURES

- Light and compact version
- Flow velocity range: 0-12 m/s, precise in low flow applications
- IDF Tri-Clamp or Union screw Connection
- Excellent for high pressure application
- Protection class: IP68 available; sensor submersible
- FEP liner is suitable for vacuum tube
- High accuracy of  $\pm 0.4\%$  value of reading (or  $\pm 0.2\%$  value of reading)
- With Forward / Reverse flowrate measurement function

## STANDARD SPECIFICATION

- |                   |                                                        |                        |                                                 |
|-------------------|--------------------------------------------------------|------------------------|-------------------------------------------------|
| ● Size            | : 10,15, 20, 25, 32, 40, 50, 65, 80 mm                 | ● Cable Entry          | : Standard: M20 Option: 1/2" NPTF               |
|                   | : 3/8", 1/2", 3/4", 1", 1-1/4", 1-1/2", 2", 2-1/2", 3" | ● Ambient Temperature  | : -25 ~ 60 °C                                   |
| ● Measuring Range | : Velocity 0-0.25 m/s Min.                             | ● Process Connection   | : IDF Tri-Clamp                                 |
|                   | 0-12 m/s Max.                                          |                        | : Union Screw                                   |
| ● Material        |                                                        | ● Grounding Resistance | : Must be less than 10 $\Omega$                 |
| Tube & Connection | : Stainless Steel 304 (Standard)                       | ● Accuracy             | : $\pm 0.5\%$ reading (Velocity $\geq 0.5$ m/s) |
|                   | : Stainless Steel 316L (Option)                        |                        | : $\pm 0.2\%$ reading                           |
| Coil Housing      | : Stainless Steel 304 (Standard)                       |                        | (Suitable for AMC3200 / 3200E converter)        |
|                   | : Stainless Steel 316L (Option)                        |                        | : $\pm 0.0025$ m/s (Velocity $< 0.5$ m/s)       |
| Liner             | : FEP                                                  | ● Temperature          | : -40 ~ 140 °C                                  |
| Electrode         | : Stainless Steel 316L                                 | ● Max. Pressure        | : 10 kgf/cm <sup>2</sup> (Standard)             |
|                   | : Hastelloy B                                          |                        | : 40 kgf/cm <sup>2</sup> (Option)               |
|                   | : Hastelloy C                                          | ● Protection Class     | : Junction box IP67, Body IP68                  |
|                   | : Titanium                                             |                        | : IP68 (Submersible)                            |
|                   | : Tantalum                                             | ● Explosion Proof      | : Ex d ia [ia] q IIC T6                         |
|                   | : Tungsten Carbide                                     |                        | (with AMC3200, AMC3200DF only)                  |
|                   | : Platinum                                             |                        |                                                 |
| ● Conductivity    | : More than 5 $\mu$ S/cm                               |                        |                                                 |

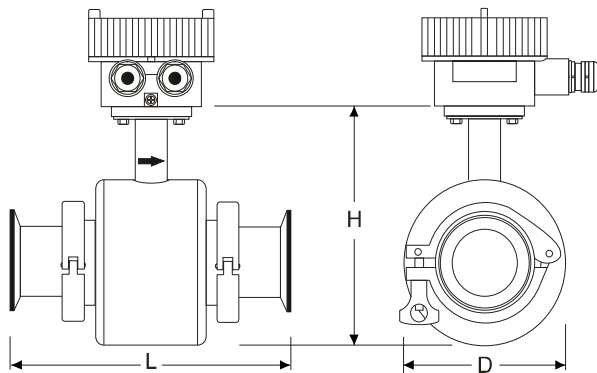


## FLOW RANGE

Unit: m<sup>3</sup>/hr

| Normal Size |        | Flow Range & Velocity Table |         |         |         |         |          |                  |
|-------------|--------|-----------------------------|---------|---------|---------|---------|----------|------------------|
| mm          | Inch   | Min.<br>0-0.25 m/s          | 1.0 m/s | 2.0 m/s | 3.0 m/s | 5.0 m/s | 10.0 m/s | Max.<br>0-12 m/s |
| 10          | 3/8"   | 0 to 0.07                   | 0.28    | 0.57    | 0.85    | 1.4     | 2.8      | 0 to 3.4         |
| 15          | 1/2"   | 0 to 0.16                   | 0.64    | 1.27    | 1.91    | 3.18    | 6.4      | 0 to 7.6         |
| 20          | 3/4"   | 0 to 0.28                   | 1.13    | 2.26    | 3.39    | 5.65    | 11.3     | 0 to 13.6        |
| 25          | 1"     | 0 to 0.44                   | 1.77    | 3.53    | 5.30    | 8.84    | 17.7     | 0 to 21.2        |
| 32          | 1-1/4" | 0 to 0.72                   | 2.90    | 5.79    | 8.69    | 14.5    | 29.0     | 0 to 34.7        |
| 40          | 1-1/2" | 0 to 1.13                   | 4.52    | 9.05    | 13.6    | 22.6    | 45.2     | 0 to 54.3        |
| 50          | 2"     | 0 to 1.77                   | 7.07    | 14.1    | 21.2    | 35.3    | 70.7     | 0 to 84.8        |
| 65          | 2-1/2" | 0 to 2.99                   | 11.9    | 23.9    | 35.8    | 59.7    | 119      | 0 to 143         |
| 80          | 3"     | 0 to 4.52                   | 18.1    | 36.2    | 54.3    | 90.5    | 181      | 0 to 217         |

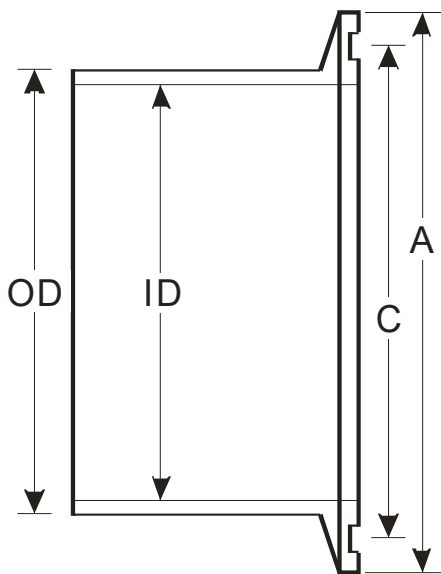
## DIMENSIONS



| Size |        | Dimensions (mm) |     |     | Weight |
|------|--------|-----------------|-----|-----|--------|
| mm   | Inch   | L               | D   | H   | kg     |
| 10   | 3/8"   | 150             | 84  | 132 | 2.1    |
| 15   | 1/2"   | 150             | 84  | 132 | 2.1    |
| 20   | 3/4"   | 150             | 84  | 132 | 2.3    |
| 25   | 1"     | 150             | 99  | 147 | 2.7    |
| 32   | 1-1/4" | 150             | 99  | 147 | 3.0    |
| 40   | 1-1/2" | 150             | 99  | 147 | 4.0    |
| 50   | 2"     | 150             | 112 | 160 | 4.8    |
| 65   | 2-1/2" | 200             | 120 | 168 | 5.8    |
| 80   | 3"     | 200             | 137 | 185 | 7.8    |

## TRI-CLAMP SPECIFICATIONS

Unit: mm

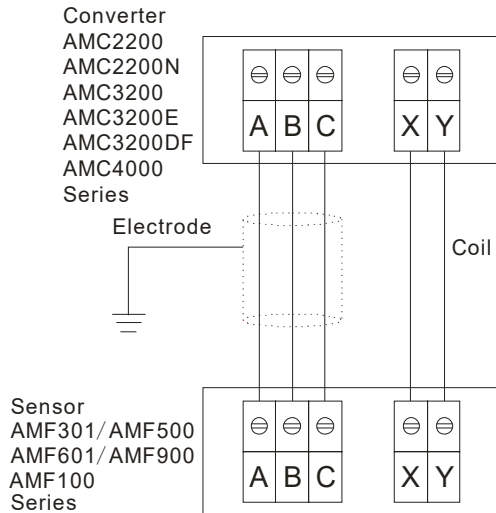


| HAB       |  |  |  |      |  |        |      |        |      |
|-----------|--|--|--|------|--|--------|------|--------|------|
| Pipe Size |  |  |  | 1"   |  | 1-1/2" | 2"   | 2-1/2" | 3"   |
|           |  |  |  | 25   |  | 40     | 50   | 65     | 80   |
| ID        |  |  |  | 22.1 |  | 34.8   | 47.5 | 60.2   | 72.9 |
| OD        |  |  |  | 25.4 |  | 38.1   | 50.8 | 63.5   | 76.2 |
| A         |  |  |  | 50.4 |  | 50.4   | 64.0 | 77.4   | 91.0 |
| C         |  |  |  | 43.6 |  | 43.6   | 56.3 | 70.6   | 83.3 |

| HDB (DIN 32676) |      |      |      |      |        |        |      |        |      |
|-----------------|------|------|------|------|--------|--------|------|--------|------|
| Pipe Size       | 3/8" | 1/2" | 3/4" | 1"   | 1-1/4" | 1-1/2" | 2"   | 2-1/2" | 3"   |
|                 | 10   | 15   | 20   | 25   | 32     | 40     | 50   | 65     | 80   |
| ID              | 10.0 | 16.0 | 20.0 | 26.0 | 32.0   | 38.0   | 50.0 | 66.0   | 81.0 |
| OD              | 14.0 | 20.0 | 24.0 | 30.0 | 36.0   | 42.0   | 54.0 | 70.0   | 85.0 |
| A               | 34.0 | 34.0 | 34.0 | 50.5 | 50.5   | 50.5   | 64.0 | 91.0   | 106  |
| C               | 27.5 | 27.5 | 27.5 | 43.5 | 43.5   | 43.5   | 56.5 | 83.5   | 97.0 |

| HKB (ISO 2852) |      |      |      |      |        |        |      |        |      |
|----------------|------|------|------|------|--------|--------|------|--------|------|
| Pipe Size      | 3/8" | 1/2" | 3/4" | 1"   | 1-1/4" | 1-1/2" | 2"   | 2-1/2" | 3"   |
|                | 10   | 15   | 20   | 25   | 32     | 40     | 50   | 65     | 80   |
| ID             | 12.0 | 17.2 | 21.3 | 22.6 | 31.3   | 35.6   | 48.6 | 60.3   | 72.9 |
| OD             | 16.0 | 21.2 | 25.3 | 25.6 | 34.3   | 38.6   | 51.6 | 64.1   | 76.7 |
| A              | 34.0 | 34.0 | 34.0 | 50.5 | 50.5   | 50.5   | 64.0 | 77.5   | 91.0 |
| C              | 27.5 | 27.5 | 27.5 | 43.5 | 43.5   | 43.5   | 56.5 | 70.5   | 83.5 |

## WIRING DIAGRAM



## MODEL SELECTION GUIDE

| AMF601 Series                   |   |         |    |   |         |                                                                       |     |                                           |  |  |
|---------------------------------|---|---------|----|---|---------|-----------------------------------------------------------------------|-----|-------------------------------------------|--|--|
| Example: AMF601-F050-SK4-030-EX |   |         |    |   |         |                                                                       |     |                                           |  |  |
| AMF601-                         | X | XXX     | -X | X | X-      | XXX                                                                   | -XX | Description                               |  |  |
| Liner                           | F |         |    |   |         |                                                                       |     | FEP                                       |  |  |
| Size                            |   | 010-080 |    |   |         |                                                                       |     | 10-80 mm                                  |  |  |
| Electrode Material              |   |         | -S |   |         |                                                                       |     | Stainless Steel 316L                      |  |  |
|                                 |   |         | -T |   |         |                                                                       |     | Titanium                                  |  |  |
|                                 |   |         | -B |   |         |                                                                       |     | Hastelloy B                               |  |  |
|                                 |   |         | -C |   |         |                                                                       |     | Hastelloy C                               |  |  |
|                                 |   |         | -A |   |         |                                                                       |     | Tantalum                                  |  |  |
|                                 |   |         | -P |   |         |                                                                       |     | Platinum                                  |  |  |
|                                 |   |         | -U |   |         |                                                                       |     | Tungsten Carbide                          |  |  |
|                                 |   |         | -Z |   |         |                                                                       |     | Others                                    |  |  |
| Process Connection              |   |         | A  |   |         |                                                                       |     | Tri-Clamps, HAB                           |  |  |
|                                 |   |         | D  |   |         |                                                                       |     | Tri-Clamps, HDB (DIN 32676)               |  |  |
|                                 |   |         | K  |   |         |                                                                       |     | Tri-Clamps, HKB (ISO 2852)                |  |  |
|                                 |   |         | S  |   |         |                                                                       |     | Screw Unions (DIN 11851)                  |  |  |
| Connections / Housing Material  |   |         | 4- |   |         |                                                                       |     | S.S. 304 Connections / S.S. 304 Housing   |  |  |
|                                 |   |         | A- |   |         |                                                                       |     | S.S. 316L Connections / S.S. 316L Housing |  |  |
| Installation & Cable Length     |   |         |    |   | NNN     |                                                                       |     | Compact Version                           |  |  |
|                                 |   |         |    |   | 005-300 |                                                                       |     | Remote Version, Cable 5-300 m             |  |  |
| Option                          |   |         |    |   | -NN     | None                                                                  |     |                                           |  |  |
|                                 |   |         |    |   | -EX     | Explosion Proof, Ex d ia [ia] q IIC T6 (with AMC3200, AMC3200DF Only) |     |                                           |  |  |
|                                 |   |         |    |   | -HP     | Pressure Higher Than 10 kgf/cm <sup>2</sup>                           |     |                                           |  |  |
|                                 |   |         |    |   | -PT     | Cable Entry 1/2" NPTF                                                 |     |                                           |  |  |
|                                 |   |         |    |   | -LR     | Low-resistance coil (with AMC3200DF only)                             |     |                                           |  |  |

**BUILD OPTIONS-(COMPACT OR REMOTE)**

